



Yusuf Erdem

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SUMMARY

I'm a curious and driven Computer Engineering student at Ankara Yildirim Beyazit University, with a broad interest in technology and a passion for learning. I've twice participated in the Work and Travel program, spending a total of eight months living and working in the United States. During this time, I thrived in diverse, **multicultural environments** connecting with people from all over the world and gaining valuable insights into different cultures and ways of thinking. These experiences enhanced my **adaptability**, **communication skills**, and global perspective. I'm also a certified voice actor and a language enthusiast, always looking for ways to combine creativity with technical problem-solving. I enjoy stepping out of my comfort zone, exploring new ideas, and contributing meaningfully to innovative teams.

EDUCATION

Bachelor of Science in Computer Engineering
Ankara Yildirim Beyazit University (AYBU) — Ankara, Turkey
Expected Graduation: 2026 | GPA: 3.1 / 4.0

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Systems, Software Engineering, Computer Networks, NLP, Probability & Statistics, Neural Networks

EXPERIENCE

Data Science Intern, Acun Medya Akademi January 2025-March 2025, Remote

- Participated in data science projects analyzing and interpreting large-scale datasets, gaining practical experience in real-world ML workflows
- Extracted actionable insights using Python, SQL, and data visualization libraries (Pandas, Matplotlib, Seaborn)
- Applied machine learning techniques including data preprocessing, feature engineering, and statistical analysis methods to support business decisions

Software Developer, Belsis (Post-Internship) September 2023-April 2024, Ankara, Turkey

- Engineered custom Odoo ERP modules for inventory management, sales tracking, and business intelligence reporting using Python and the Odoo framework
- Deployed production system supporting 50+ concurrent users across the organization, reducing manual reporting time by automating workflows
- Managed and optimized PostgreSQL relational databases using pgAdmin and DBeaver, ensuring data integrity and query performance
- Streamlined team collaboration using Git, GitLab, and Jira for version control and project tracking

Software Developer Intern, Belsis June 2023-September 2023, Ankara, Turkey

- Designed and developed a parcel selection GIS application using OpenLayers, contributing to a municipal project adopted by multiple Turkish municipalities
- Architected the backend with C#, ASP.NET Core, and SQL, enabling efficient spatial database operations via PostgreSQL + PostGIS
- Developed and optimized the frontend using Node.js, JavaScript, HTML, and CSS, delivering a responsive and user-friendly mapping interface

PROJECTS

AI-Based Health Risk Scoring & Resource Allocation | (Capstone) Graduation Project | 2025 – 2026

- Participated a 3-person team building a disease prediction system for diabetes, hypertension, and stroke **using non-laboratory patient data**
- Designed a **two-layer ML architecture**: classification layer for disease identification, and risk scoring layer for severity assessment
- Selected 82% accurate Logistic Regression over 98% overfitted XGBoost**, prioritizing **clinical validity and generalizability** over raw accuracy
- Solved class imbalance using **SMOTE oversampling** and applied comprehensive feature engineering for improved model robustness
- Tech: Python, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

End-to-End Crypto Forecaster (Live in Production) | Personal Project | 2025 – Present

- Built and deployed a full MLOps pipeline: Dockerized PostgreSQL data lake, idempotent ETL, automated feature engineering, and model retraining
- Compared **SARIMAX** (1.8% MAPE) vs **Bi-LSTM** (3.2% MAPE), proving statistical models outperform deep learning on high-noise cryptocurrency data
- Deployed a live Streamlit dashboard for **real-time BTC price forecasting**, accessible at btcforecaster.streamlit.app
- Implemented CI/CD via GitHub Actions for **automated weekly model retraining**
- Tech: Python, TensorFlow, scikit-learn, PostgreSQL, Docker, FastAPI, Streamlit

IoT Intrusion Detection System | IoT Course Project | 2025

- Developed an XGBoost-based intrusion detection system for IoT networks, **achieving 99% precision and 0.97 F1-score on the EdgeIoT dataset**
- Built a real-time Flask **dashboard for live network traffic monitoring** with explainable alerts and color-coded threat level visualization
- Performed extensive **feature selection** and **hyperparameter tuning** to maximize detection accuracy while minimizing false positives
- Tech: Python, XGBoost, Flask, scikit-learn, Pandas

Azerbaijani Sentiment Analysis | NLP Course Project | 2025

- Built a **FastText + GRU sentiment classifier** for Azerbaijani YouTube comments, a **low-resource language** with limited NLP tooling
- Curated a dataset of **36,000 comments across 5 domains** and developed a custom language filter achieving 85.7% noise removal accuracy
- **Achieved Macro-F1 score of 0.8016**; demonstrated that frozen embeddings outperform fine-tuned embeddings for low-resource sentiment tasks
- Tech: Python, FastText, PyTorch, NLP, Custom Text Filtering

ROTASYON: Smart Urban Mobility System | UDHAM National Competition | Team of 5 | 2025

- **Won "Most Ecological Project" award** at the 2025 UDHAM National Universities Idea Competition; received **50,000 TL grant**
- Designed backend API for **congestion pricing and low-emission zone** management with intelligent vehicle routing for 5+ Turkish cities
- Designed vehicle eligibility logic, route optimization algorithms
- Tech: Python, FastAPI, PostgreSQL, Google Maps API, Mobile (iOS/Android)

SKILLS

- Machine Learning: XGBoost, TensorFlow, scikit-learn, PyTorch, Feature Engineering
- Data Engineering: PostgreSQL, Docker, ETL Pipelines, GitHub Actions CI/CD
- Backend: FastAPI, Flask, ASP.NET Core, RESTful APIs
- NLP: FastText, Sentiment Analysis, Text Classification

CERTIFICATIONS

- IELTS Academic — Band 8 (C1 Proficient Level)
- Complete A.I. & Machine Learning, Data Science Bootcamp — Udemy (2025)

AWARDS

- "Most Ecological Project" Award — 2024 UDHAM National Universities Idea Competition | 50,000 TL grant recipient

LANGUAGES

- Turkish: Native
- English: **Full Professional Proficiency (IELTS Band 8)**
- German B1